JAVA PROGRAMMING PRACTICES

File Handling: Count pe and pi

Basic Information

Week	Week 10
Source file	PatternFrequency.java
Concepts	File reading, String methods, try-catch, IOException.

Objective / Aim

To implement file handling: count pe and pi using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Read sample.txt.
2. Scan adjacent character pairs and count pe and pi.
3. Display both counts.

UML / Program Structure

```
PatternFrequency
Reads sample.txt; counts pe and pi
```

Source Code

```
import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.Paths;
import java.nio.charset.StandardCharsets;
public class PatternFrequency
{
    public static void main(String[] args)
    {
        try
        {
            String text = new String(Files.readAllBytes(Paths.get("sample.txt")),
            StandardCharsets.UTF_8).toLowerCase();
            int pe = 0, pi = 0;
            for (int i = 0; i < text.length() - 1; i++)
            {
                String pair = text.substring(i, i + 2);
                if (pair.equals("pe")) pe++;
                if (pair.equals("pi")) pi++;
            }
            System.out.println("'pe' - no of occurrences - " + pe);
            System.out.println("'pi' - no of occurrences - " + pi);
        }
        catch (IOException e)
        {
            System.out.println("Could not read sample.txt");
        }
    }
}
```

Compilation and Execution

```
javac PatternFrequency.java
java PatternFrequency
```

Output Screen

```
'pe' - no of occurrences - 20
'pi' - no of occurrences - 12
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.